

ED Research System — Handoff (Freeze Snapshot)

Timestamp: 2025-08-24 18:58 UTC

Status: Phase 1 (Ops) — COMPLETE; Phase 2 (Clinical Integration) — COMPLETE.

Purpose: Freeze current working state for safe transfer; ML remains advice-only.

Architecture & Invariants

Core: WorkflowState + TinyCritics; TrackerService (+ repos, QR, SOP); Streamlit UI (sp only MLP bridge; Gate engine wired).

Invariants: CONFIG frozen (only MODEL_BUNDLE_PATH may change); CSV/dir semantics; QR generate+manual-fallback; SOP auto-pull offline-safe; adjustable thresholds (equipment vitals overdue).

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Implemented (Evidence)

Tracker: moves logging; robust EquipmentRepository.read; movement analytics.
QR: generate + manual fallback update path.
SOPs: refresh_sop_registry() present; offline-safe summary.
UI: split helpers; adjustable thresholds present.
ML: bundle autodetect + load; advice_gatepos_mlp.csv written; calibrator-safe.
Gate Engine: TTL_MAX_OK, GATES_WIRED_OK, PHASE2_BUNDLE_SMOKE_OK are OK.

How to Run (Safe Defaults)

Do not mutate frozen CONFIG; use overlay CFG for tests.

Bundle: `/kaggle/input/ed-pipeline-bundle-ui/ed_phase2_model_thr_patched(1).joblib`

MIMIC-IV-ED demo: `/kaggle/input/mimic-iv-demo-v2-2/*.csv`

Paths (current run)

EQUIPMENT_STATUS_PATH: /kaggle/working/equipment_status.csv
EQUIPMENT_MOVES_LOG_PATH: /kaggle/working/moves_log.csv
SOP_REGISTRY_PATH: /kaggle/working/sop_registry.csv
QR_OUTPUT_DIR: /kaggle/working/qrs
EVENT_LOG_PATH: /kaggle/working/event_log.jsonl
MODEL_BUNDLE_PATH: autodetected in /kaggle/working/*.joblib
Advice outputs: /kaggle/working/advice_gatepos_mlp.csv

Known Limits & Next Steps

Limits: demo subset; sklearn version warnings (acceptable); UI launch depends on Streamlit runtime.

Next: labels + multi-task MLP (advice-only) as refiners; per-head calibration; optional telemetry.

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